A course advisory dataset for IIT Gandhinagar M.Tech students

CS 203 Assignment 1, Project 4

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1 Introduction

We built a dataset for a small language model that answers three kinds of M.Tech advising questions: what is still left for my degree, can I register for a course next term, and which electives should I take. The scope is four programmes in the CS and EE departments, MTech-CSE, MTech-AI, MTech-EE and MTech-ICDT, each with a thesis and a coursework track [E1]. A rule engine built from the institute's documents computes every gold answer (104 courses, 77 rules, 180 synthetic students) [E2, E3, E4], a local LLM (Qwen2.5-1.5B-Instruct) paraphrases questions and justifications, and two people audit the paraphrases.

600	1526	80	76	19
advisory pairs, 200 per intent [E5]	verified augmented pairs [E6]	doubly labelled audit items [E7]	frozen gold test items [E8]	real form questions (shifted set) [E9]

Every number here is copied from a repository file; Appendix A lists them as E1, E2, …, cited in square brackets.

2 Part A: Collection and instrumentation

104	77	22	3083	0
courses [E2]	degree rules, 20 of them EE [E3]	course plan files [E15]	metadata rows, one per sample [E11]	enforceable PG prerequisites [E19]

2.1 Collection pipeline. The raw layer holds eight academic advisories, the revised CSE M.Tech course structure 2025-26, the Sem-I 2026-27 timetable and four pages of the EE M.Tech site, each with a sha256 and kept untouched [E10]. `metadata.csv` has one row per sample with its source, timestamp (`collected_at_utc`), collector, tool and tool version and content hash [E92]: 3083 rows (578 raw, 2505 processed), covering source files, pages, timetable rows, plans, catalog records, every rule with a stated value (76 of 77), 180 students, 600 pairs, 1526 augmented children and 19 form responses [E11, E12].

Course plans. For the 50 CS and ES codes, 15 plans downloaded, 12 links need an IITGN login (not downloaded) and 23 codes have no link [E13]; for the 25 EE codes the split is 7, 4 and 14 [E14], so 22 plan files were saved [E15]. *Curriculum JSON.* The raw catalog keeps 136 records with their defects (duplicates, the "Algrithms" typo, float credits) [E16]. Cleaning turns them into 104 courses (67 basket, 15 open elective, 6 stream core, 2 compulsory, 6 requirement, 8 UG background) and logs all 78 changes [E17, E18]. Every course and rule carries its source document with page, row or section and its own hash; every rule stores a verbatim quote that the build checks against the page. One rule, maximum credits per semester, is marked not stated [E3]. *Prerequisite graph.* 103 of 104 courses state no prerequisite; CS 613 has a manual entry with an empty list, because its plan says "Prerequisite (Optional)" with UG courses "or equivalent" (proposed by the pipeline assistant, confirmed by hem436) [E19, E20]. The engine therefore has two tiers: hard prerequisites (checked, empty in practice) and recommended background, which 3 courses carry [E21]. The graph has 104 nodes, 9 edges (all recommended), no missing codes and no cycles [E22]. D4 stores UG codes from plans as recommended background, and D5 (hem436) says they never warn an M.Tech student [E23]. *Rule engine and synthetic transcripts.* The engine generates 180 students from seed 203: CSE 64, EE 51, AI 41, ICDT 24; 92 thesis and 88 coursework; 1422 transcript entries with 0 rule violations [E4, E24].

2.2 Consent and licensing. The institute documents are public regulations, used only for non-commercial coursework and not redistributed; course plans were saved only where the link opened without a login [E93]. Form respondents tick a required consent box; the consent register has one row per response with consent

and form version [E94]. No names or emails are collected, the Google timestamp is dropped and contact details in free text are removed.

2.3 Data flow. Figure 1 shows how data moves through the pipeline; each arrow is a DVC stage [E95]. The collection log records the fetch outcomes, tool versions and defects found [E96].

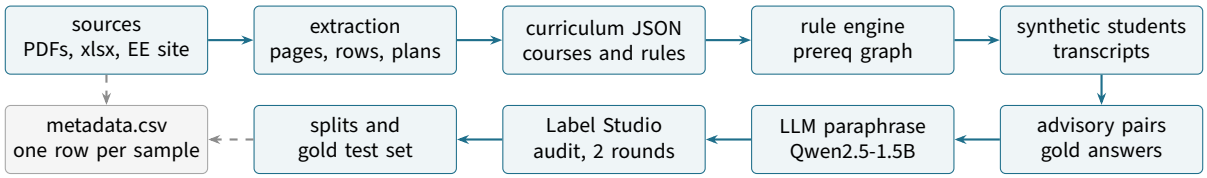


Figure 1: Data flow from the sources to the gold set. Dashed: every layer also writes its rows to `metadata.csv` [E95, E11].

3 Part B: Data validation

25	7 of 10	24 + 1	15 of 15	2
checks: Pandera, networkx, rapidfuzz, rule engine [E25]	raw-catalog checks fail [E26]	processed: pass + one warning [E27]	injected faults caught [E29]	gold bugs found by people [E33, E34]

3.1 Validation suite. 25 checks run on the raw catalog, the processed data and a corrupted copy, using Pandera schemas, networkx graph checks, rapidfuzz title matching and the rule engine itself [E25]. They cover the Project 4 checks: prerequisites exist (V11), no cycles (V12), credit totals (V13), no duplicate codes (V06), transcripts pass the engine (V17 to V20) and no student or template in both train and test (V25) [E97].

3.2 Before and after. On the raw catalog 7 of 10 applicable checks fail; on processed data 24 pass and V10 warns about two different courses with near-identical titles, kept on purpose (Figure 2) [E26, E27].

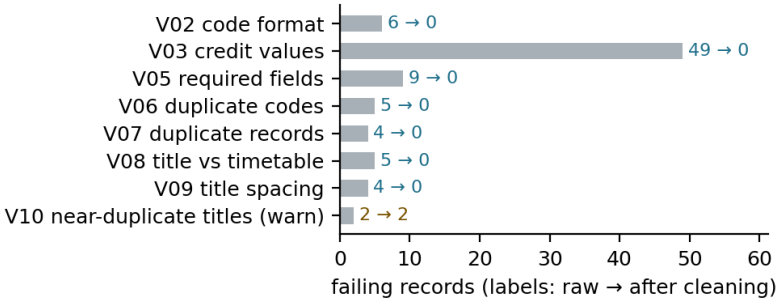


Figure 2: Failing records per check before and after cleaning [E28]. Full table: Table 1.

3.3 Corrupted data. A DVC stage injects 15 faults with seed 209, among them a prerequisite cycle, an unknown prerequisite, total credits changed from 82 to 80, a below-600 cap breach and train/test leakage; all 15 are caught [E29, E32, E30]. One planned fault, a per-semester credit cap, was not injected because no source states a cap [E31].

3.4 What human review added. While labelling the first item, hem436 saw “Other baskets (): 0 credits earned” for an MTech-AI student: the engine read other baskets from the student’s own one-stream programme. The fix follows the CSE structure and changed 100 of 600 gold answers, all MTech-AI [E33]. Later hem436 noticed answers telling M.Tech students they lacked IITGN UG courses; D5 removed these warnings and changed 53 gold answers and 0 query texts [E34]. Both have regression tests and came before any finished annotation. The AI fix would have changed 12 of the 180 students, so generation keeps the old basket view while gold answers use the fixed rule [E35, E36].

4 Part C: Labelling and agreement

2×40	0.38	0.35	55 / 21 / 4	76
items, both annotators [E37]	decision kappa in round 2, up from 0.23 [E41]	category kappa in round 2, up from 0.17 [E41]	accept / edit / reject [E53]	gold test items [E8]

4.1 Guidelines. Round 1 used guidelines v1. After it we analysed every disagreement and wrote v2, which adds seven rules, each tied to a round-1 item: state every gold fact, any contradicted number is a wrong fact, a clear edit-or-reject threshold, query checks, two-problem items, gold is the truth, formatting is not an error (Appendix C) [E40].

4.2 Independent annotation. A1 (hem436) and A2 (myankb08) labelled the same items in Label Studio 1.23.2, separately, in two disjoint rounds of 40 drawn from the test split with seed 208 [E37, E38, E39]. Each item shows the student, the template query, the paraphrase, the gold answer and the rewritten justification. Labelling test items is how the gold test set is built: the rounds only audit and correct the text, and no model is trained, tuned or selected on any test item. Figure 3 shows the workflow.

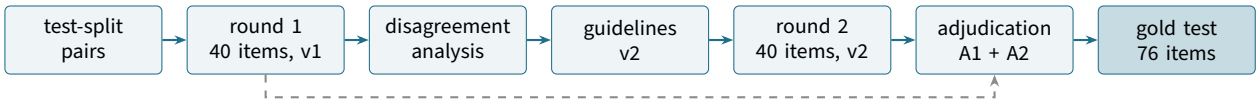


Figure 3: Labelling workflow. Both rounds feed adjudication (dashed: round 1 disagreements) [E37, E48, E8].

4.3 Label order. Three levels in a fixed order: decision (accept, edit, reject), one error category, then the edit text (Appendix C) [E98]. Deciding first keeps the category list from steering the decision, and the edit comes last, once the problem is named.

4.4 Agreement. Decision kappa went from 0.227 [0.000, 0.469] to 0.377 [0.094, 0.630], and category kappa from 0.171 [0.007, 0.353] to 0.350 [0.064, 0.620], with 2000 bootstrap resamples [E41, E42]. Raw agreement went from 65.0% to 77.5% on decision and from 62.5% to 75.0% on category; disagreements fell from 17 to 12 [E44, E45]. The CIs overlap and the rounds use different items, so the gain is in the expected direction but not conclusive. Round 1 was about strictness: A2 accepted 34 of 40 items and A1 24, and in 11 of the 17 disagreements A1 flagged an item that A2 accepted [E46, E47]. Causes: vague “almost there” justifications, wrong numbers in fluent text, the edit-or-reject threshold, and fixing correct gold from memory. In round 2 the main split left was wrong numbers in fluent text, and the paraphraser kept turning “coursework track” into “workshop track”.

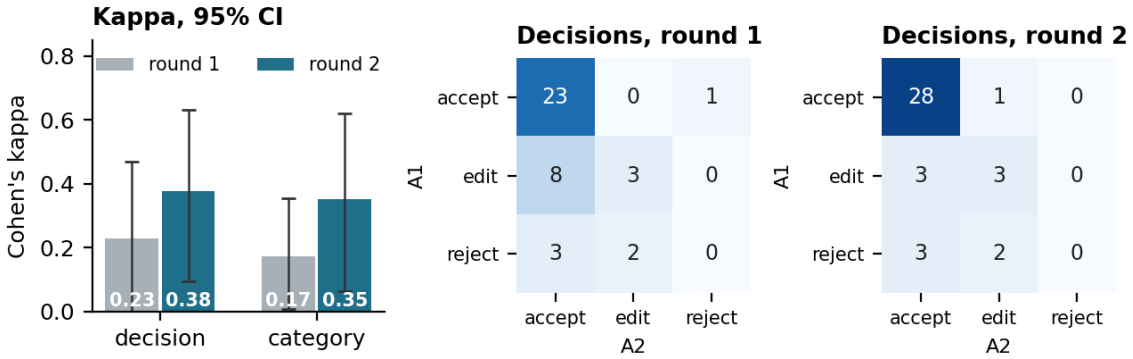


Figure 4: Left: Cohen’s kappa with bootstrap 95% CI [E41]. Middle and right: decision confusion, A1 rows and A2 columns, round 1 and round 2 [E43].

4.5 Gold test set. 50 items agreed fully; the other 30 (17 in round 1, 13 in round 2) are the 29 disagreements plus one item where both chose edit with different texts [E48]. The assistant drafted a proposal for each of

the 30 with a one-line reason [E49]; A1 and A2 reviewed the 29 disagreements together, approving 28 and overriding 1 [E50], and A1 decided the extra item [E51]. None stayed unresolved [E52]. The gold test set is the 76 accepted or edited items (27 eligibility, 27 elective recommendation, 22 remaining requirements), frozen with sha256 `cd2e08dc...` [E54, E55].

4.6 Real queries. 19 consenting responses were imported, 0 excluded; 8 came from form v1 and 11 from v2, which added three example questions after respondents were unsure what to ask [E9, E56, E57]. 9 questions were asked in person by classmates and entered by hem436 after rewording with ChatGPT; the 10 self-submitted rows keep the students' own words [E58]. They are informal ("sem 3 of m.tech EE,SPML stream") and often fall outside the three intents, so they form the out-of-distribution shifted split. By kind: 9 remaining requirements, 5 elective advice, 2 eligibility, 3 other [E59].

5 Part D: Augmentation and active learning

1526	36 of 120	232 of 336	0.43
of 1610 augmented pairs kept [E6]	LLM paraphrases kept [E60]	base-model answers wrong [E63]	failures found by top-k at budget 100; random 0.30 [E65]

5.1 Augmentation. Typo and transcript children must leave the gold answer unchanged; transcript-length children must get the correctly recomputed answer. LLM paraphrase children must also keep every fact the question states (programme, stream, track, semesters completed, number of electives, term) and must not turn the question back on the student [E61]. Most LLM paraphrases fail this: the track is lost in 47 failures (26 of them track alone), semesters completed in 35, stream in 21, and 10 turn the question to the student [E62], the same "workshop track" errors the annotators caught. The typo and transcript checks pass at 1.0 by construction, since those transforms never touch the facts the engine reads.

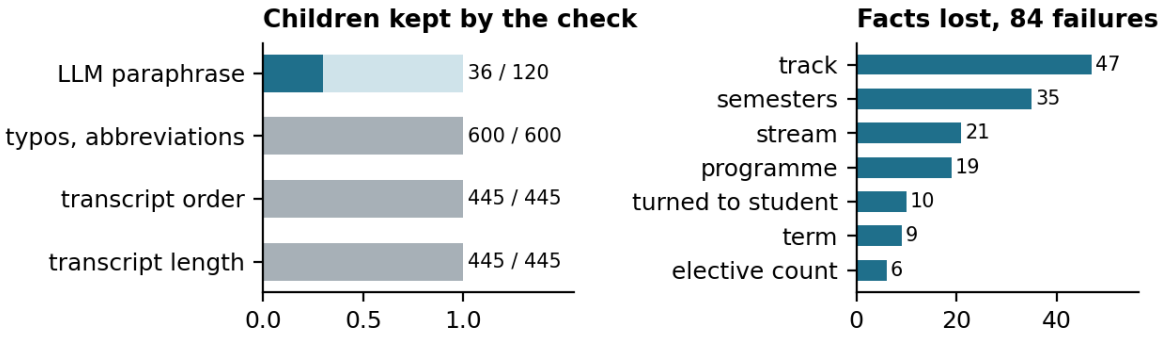


Figure 5: Left: share of children each check keeps [E60]. Right: facts lost by the failed LLM paraphrases (one can lose several) [E62]. Full table: Table 2.

5.2 Active learning. The base model (same Qwen, greedy) answered the 336 train pairs; 232 answers are wrong: wrong remaining 94, ineligible recommendation 64, wrong eligibility 56, invalid JSON 18, hallucinated code 2 [E63, E64]. Random is the mean of 5 seeds [E66]. The score and the failure test are the same comparison, so every top-k pick is a failure (precision 1 by construction); the informative part is which failures each strategy reaches [E67]. Plain top-k takes every pick from one intent and covers 0.60 of the failure types; the intent-stratified variant, added after the first run, keeps most of the gain (0.38 at budget 100) and covers all failure types and intents [E65, E68]. This is a label-efficiency comparison, not a learning curve.

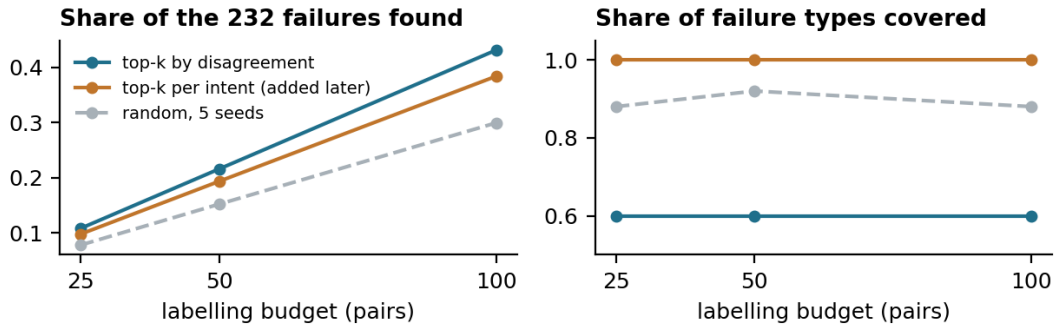


Figure 6: Left: share of the 232 base-model failures each strategy finds; right: share of failure types it reaches [E65]. Full table: Table 3.

6 Part E: Versioning and splits

26	0	7 of 22	24	19
DVC pipeline stages [E69]	students shared between splits [E77]	templates used only in test [E76]	ICDT students held out [E78]	real questions in the shifted split [E9]

6.1 Versioning. Human and raw inputs (sources, plans, form CSV, LLM cache, labelling files) are added with `dvc add`; everything else comes from the 26 stages, with EE fetching frozen [E70, E69]. The remote is a local folder next to the repo [E71]. `dvc.lock` pins every output, and a full offline `dvc repro` reproduced courses, rules, students, pairs, paraphrases, splits, labels, the gold set and base-model answers byte for byte; only the run-time column of the metadata file differs [E72, E73].

6.2 Splits. Students are assigned first, per programme (seed 207; 0.65 / 0.15 / 0.20), and each pair is then written with a template allowed in its split [E75]; one template per family is test-only. All MTech-ICDT students form a programme holdout (the smallest programme and the only one without streams), and the real form questions form the shifted split. Augmented pairs inherit their parent's split. All overlap checks are zero and no pair was dropped [E77]; eligibility is balanced in every split (17 eligible and 17 not in test) [E78].

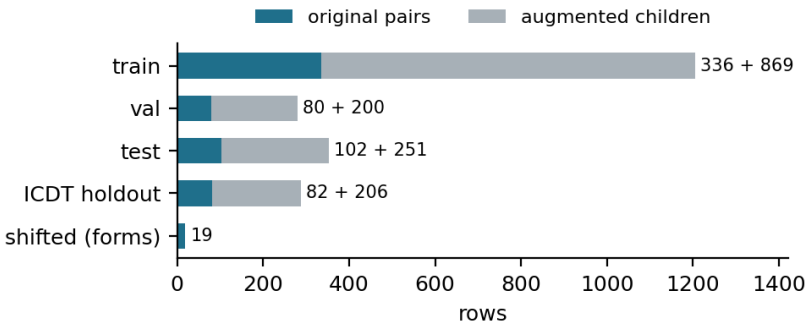


Figure 7: Rows per split: original pairs plus augmented children [E78, E9]. Full table: Table 4.

6.3 Final tag. The tag `a1-final` is set on the commit that contains this report [E74].

7 Part F: Report and reproducibility

252	8	3	3.10.12
tests pass [E79]	seeds in params.yaml [E80]	LLM backends, recorded per row [E84]	Python; DVC 3.67.1 [E81]

7.1 Dataset card. DATASET_CARD.md gives the sources, sizes (104 courses, 180 students, 600 pairs plus augmented children), balance (200 pairs per intent; 100 eligible and 100 not) and known biases: synthetic students, no enforceable prerequisites, credit defaults, a partial ICDT list, one known timetable term and a small real-query set [E99, E5, E100].

7.2 Data reproducibility. ENVIRONMENT.md records the laptop (Ubuntu 22.04.5, Ryzen 5 2500U, CPU only), Python, DVC, Label Studio, the model digest and the Kaggle GPU (Tesla T4, CUDA 12.8, torch 2.11.0+cu128, transformers 5.16.1) [E81, E82]. The pipeline prompt README lists model, versions and backends per prompt [E83]; all eight seeds live in `params.yaml` [E80]. LLM answers are replayed from the DVC-tracked cache, so a rerun gives the same text without a GPU.

7.3 Label reproducibility. LABELLING_PROTOCOL.md records who labelled what, tool versions, config and guidelines per round, label order, item order and seed, what annotators saw, the adjudication process and a timeline matched to the exports, which are never edited [E101].

7.4 Repository and README. The README maps each Part to its code and outputs and rebuilds everything with four commands [E102]:

```
python3.10 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/dvc pull && .venv/bin/dvc repro && .venv/bin/pytest -q
```

8 Limitations

- Students are synthetic and valid by construction; no source states an enforceable PG prerequisite, so eligibility mostly tests offering and already-taken rules. Kappa rests on 40 items per round, so the CIs are wide.
- LLM backends are mixed: Ollama 0.1.38 could not load the qwen2 tokenizer, so the same weights ran on llama.cpp, and part of the paraphrases ran on a Kaggle T4. Each row records its backend; outputs differ across backends [E84, E85].
- Form examples were added mid-collection, which may pull later wording toward them. We have 19 real questions, below the target of 30; 9 were reworded with ChatGPT, so only 10 keep the students' own words [E86, E58].
- Five accepted gold items still contain the paraphraser's "workshop track" error, missed by both annotators; they are listed in `HANDOFF_A2.md` so Assignment 2 can exclude them [E103].
- The EE site marks the ICDT list "to be updated soon", so ICDT answers rest on a partial list [E87].

9 Contribution table

member	commits	work
Myank Singh Beniwal (25210087; myankb08)	3	First data-loading commit; annotator A2 in both rounds (exports committed from his own account); joint adjudication with A1 (all 29 reviewed decisions are A1+A2). The docs record form distribution by hem436 only [E88, E89, E90].
Shekhar Mohanty Hement (25210122; hem436)	52	Everything else: scope decisions, pipeline and rule engine (built with Claude Code), sources, form creation and sharing, annotator A1, joint adjudication, DVC, documentation and this report.

A Evidence index

Every number in the body, with the file that backs it (paths from the repo root) and the field or row.

id	what	file and field
1 Introduction		
E1	4 programmes, thesis and coursework tracks	`docs/DECISIONS.md`, rows D1, D2, D3
E2	104 courses	`data/processed/courses.json`, list length; `supporting_files/B_engine/prereq_graph_report.json` `nodes`
E3	77 rules; 20 EE (`ee.*`); 1 not stated (max credits per semester)	`data/processed/rules.json`, `rules` keys, prefix `ee.`, `not_stated: true`
E4	180 students; CSE 64, EE 51, AI 41, ICDT 24; 92 thesis, 88 coursework; 1422 entries; 0 violations	`supporting_files/B_engine/transcripts_summary.json`, `n_students`, `programmes`, `tracks`, `transcript_entries`, `violations`
E5	600 pairs, 200 per intent	`supporting_files/B_engine/queries_summary.json`, `n_pairs`, `per_intent`
E6	1526 children kept of 1610	`supporting_files/E_splits/split_summary.json` `n_children_in` (sum of `child_rows_per_split`); `supporting_files/D_augmentation/verify_report.json` `children_in`
E7	80 audit items (40 per round)	`data/labels/audited.jsonl`, row count by `round`
E8	76 gold test items	`supporting_files/E_splits/gold_summary.json`, `n_gold`
E9	19 real form questions	`supporting_files/C_labelling/forms_import_log.json` `rows_kept`; `supporting_files/E_splits/shifted_summary.json` `n_rows`
2 Part A: Collection and instrumentation		
E10	8 advisories, CSE structure, timetable, 4 EE pages, each with sha256	`data/raw/sources/source_metadata.json` `sources[].sha256`; `data/metadata.csv` rows with `artifact` under `data/raw/sources/`
E11	3083 metadata rows (578 raw, 2505 processed)	`data/metadata.csv`, row count and `layer` counts
E12	180 / 600 / 1526 / 19 / 76 rows for students, pairs, children, forms, rules (76 of 77)	`data/metadata.csv`, `sample_id` prefixes `student`, `pair`, `aug`, `form`, `rule`
E13	50 CS/ES codes: 15 downloaded, 12 login, 23 no link	`supporting_files/A_collection/course_plan_fetch_log.csv`, last `outcome` per `code`
E14	25 EE codes: 7, 4, 14	`supporting_files/A_collection/ee_course_plan_fetch_log.csv`, same
E15	22 plan files (15 + 7)	`data/raw/course_plans/`, file count
E16	136 raw catalog records	`data/metadata.csv` rows with `artifact` = `data/raw/catalog_raw.json`; `supporting_files/A_collection/collection_log.md` section 7
E17	67 / 15 / 6 / 2 / 6 / 8 courses by category	`data/processed/courses.json`, `category`
E18	78 cleaning changes	`supporting_files/A_collection/cleaning_changes.csv`, data rows
E19	103 none stated, 1 manual; 0 enforceable	`data/processed/courses.json`, `prereq_source` and empty `prerequisites`
E20	CS 613 manual decision	`data/raw/prereq_manual.csv`, `decided_by`
E21	3 courses with recommended background	`data/processed/courses.json`, non-empty `recommended_background`
E22	104 nodes, 9 recommended edges, 0 missing, 0 cycles	`supporting_files/B_engine/prereq_graph_report.json`, `nodes`, `edges_by_type`, `missing_prereq_codes`, `cycles`
E23	decisions D4, D5	`docs/DECISIONS.md`, rows D4, D5
E24	seed 203	`params.yaml`, `seeds.transcripts`
3 Part B: Data validation		
E25	25 checks, V01 to V25	`supporting_files/B_validation/processed/results.csv`, 25 rows; libraries in `requirements.txt`
E26	raw: 10 checks, 7 fail, 2 pass, 1 warn	`supporting_files/B_validation/raw/results.csv`, `status`

id	what	file and field
E27	processed: 24 pass, 1 warn (V10, 2)	`supporting_files/B_validation/processed/results.csv`, `status`, `n_failures`
E28	Figure 2 and Table 1 values	`supporting_files/B_validation/before_after.md`, table rows
E29	15 of 15 faults caught	`supporting_files/B_validation/corrupted/faults_caught.json`, `caught`
E30	total credits 82 to 80	same, F05 `description`
E31	1 fault not injected	`data/interim/corrupted/faults.json` (output of DVC stage `validate_corrupted`), `not_injected`
E32	seed 209	`params.yaml`, `seeds.validation_corruption`
E33	100 of 600 gold answers changed (AI fix)	`supporting_files/B_engine/gold_fix_mtech_ai_baskets.md`, "Effect"
E34	53 gold answers, 0 query texts changed (D5)	`supporting_files/B_engine/gold_fix_ug_background.md`, "Effect"
E35	12 of 180 students the AI fix would change	`supporting_files/B_engine/gold_fix_mtech_ai_baskets.md`, last section
E36	old basket view kept for generation	`src/engine/rule_engine.py`, `role_of` docstring
4 Part C: Labelling and agreement		
E37	40 items per round, 2 rounds	`supporting_files/C_labelling/round1/agreement_summary.json` and `round2/`, `n_items_compared`
E38	seed 208	`params.yaml`, `seeds.annotation_order`
E39	Label Studio 1.23.2	`ENVIRONMENT.md`, Python and tools table
E40	seven v2 rules (R1 to R7)	`supporting_files/C_labelling/round1/disagreement_analysis.md`, "Changes for guidelines v2"
E41	kappa 0.227, 0.171, 0.377, 0.350 and their 95% CIs	`supporting_files/C_labelling/kappa_ci.json`, `round*_decision`, `round*_error_category`
E42	2000 resamples	same, `n_boot`
E43	confusion matrices	`supporting_files/C_labelling/round1/confusion_decision.csv` and `round2/`
E44	raw agreement 65.0 / 62.5 and 77.5 / 75.0%	`supporting_files/C_labelling/round1/agreement_summary.json` and `round2/`, `percent_agreement`
E45	17 and 12 disagreements	same, `n_disagreements`
E46	A2 accepted 34, A1 24 (round 1)	`supporting_files/C_labelling/round1/agreement_summary.json`, `label_counts`
E47	11 of 17: A1 flagged, A2 accepted	`supporting_files/C_labelling/round1/disagreement_analysis.md`, "The main pattern"
E48	50 agreed; 17 + 13 adjudicated	`supporting_files/C_labelling/adjudication_log.json`, `agreed`, `adjudicated`
E49	30 proposals (29 + R2-Q0196)	`supporting_files/C_labelling/adjudication_proposals.json`, keys other than `_about`
E50	28 approved, 1 overridden, all A1+A2	`supporting_files/C_labelling/adjudication_decisions_export.json`, `choice`, `decided_by`
E51	extra item decided by A1	`data/labels/audited.jsonl`, `decided_by` of R2-Q0196
E52	0 unresolved	`supporting_files/C_labelling/adjudication_log.json`, `unresolved_items`
E53	55 accept, 21 edit, 4 reject	`data/labels/audited.jsonl`, `final_decision`
E54	gold by intent 27 / 27 / 22	`supporting_files/E_splits/gold_summary.json`, `by_intent`
E55	sha256 `cd2e08dcb56f5716b06981e096c3c93fd417abab1989b5d22d45befeafbc383f`	`data/gold/GOLD_SHA256.txt`, line 1
E56	0 excluded	`supporting_files/C_labelling/forms_import_log.json`, `rows_excluded`
E57	8 v1, 11 v2	`supporting_files/E_splits/shifted_summary.json`, `by_form_version`
E58	10 self-submitted, 9 entered after ChatGPT rewording	same, `by_provenance`; `supporting_files/C_labelling/forms_log.md`
E59	9 / 5 / 2 / 3 by kind	same, `by_question_kind`
5 Part D: Augmentation and active learning		

id	what	file and field
E60	36 of 120 paraphrases; Figure 5 left and Table 2	`supporting_files/D_augmentation/invariance.csv`, all rows
E61	facts the paraphrase check compares	`src/augment/verify.py`, `slot_issues`
E62	track 47 (26 alone), semesters 35, stream 21, turned to student 10	`supporting_files/D_augmentation/invariance.csv`, `fail_reasons`, counts summed over every reason that names the slot
E63	336 pool pairs, 232 failures	`supporting_files/D_active_learning/scores_summary.json`, `n`, `failed`
E64	94 / 64 / 56 / 18 / 2 by failure type	same, `failure_types`
E65	0.43, 0.38, 0.30 at budget 100; 0.60 types; Figure 6 and Table 3	`supporting_files/D_active_learning/label_efficiency.csv`, `failures_captured`, `failure_types_covered`, `intents_covered`
E66	5 random seeds	`params.yaml`, `seeds.active_learning`
E67	precision 1 by construction; not a learning curve	`supporting_files/D_active_learning/summary.md`, "Reading the first column"
E68	stratified variant added later	same, paragraph after it
6 Part E: Versioning and splits		
E69	26 stages, EE fetch frozen	`dvc.yaml`, `stages`, `fetch_ee_pages: frozen`
E70	inputs added with dvc add	`data/raw/*.dvc`, `data/raw/forms/responses_raw.csv.dvc`, `data/interim/llm_cache.jsonl.dvc`, `data/labels/round*/*.dvc`
E71	local remote `../dvc_remote`	`.dvc/config` `remote "localremote"`; `README.md`
E72	every output pinned	`dvc.lock`, stage `outs` hashes
E73	byte-for-byte rerun	`supporting_files/F_reproducibility/repro_check.txt`, `sha256sum -c` lines, `dvc status`
E74	tag a1-final	`HANDOFF_A2.md`, first paragraph
E75	seed 207; 0.65 / 0.15 / 0.20	`supporting_files/E_splits/split_summary.json`, `seed`, `fractions`
E76	7 of 22 templates test-only	same, `held_out_templates`; `supporting_files/B_engine/queries_summary.json` `per_template`
E77	all overlaps 0, 0 dropped	`supporting_files/E_splits/split_summary.json`, `overlap_checks`, `all_overlaps_zero`, `n_dropped_pairs`
E78	Figure 7 and Table 4 values; 24 ICDT students; 17 / 17 in test	same, `student_assignment`, `original_rows_per_split`, `child_rows_per_split`, `rows_per_split`, `eligibility_by_split`, `original_rows_by_programme`
7 to 9 Part F, limitations, contribution table		
E79	252 tests pass	`supporting_files/F_reproducibility/pytest.txt`, last line
E80	8 seeds	`params.yaml`, `seeds`
E81	Ubuntu 22.04.5, Ryzen 5 2500U, Python 3.10.12, DVC 3.67.1	`ENVIRONMENT.md`, Machine and Python and tools tables
E82	Tesla T4, CUDA 12.8, torch 2.11.0+cu128, transformers 5.16.1	`ENVIRONMENT.md`, GPU row
E83	model, versions, backends per prompt	`prompts/pipeline/README.md`, Model and Prompts
E84	3 backends; Ollama 0.1.38, llama.cpp, Kaggle T4	`supporting_files/D_augmentation/llm_backend.md`, sections 1, 2, 4
E85	backend per row	`data/processed/queries_para.jsonl`, `para_backend`
E86	target of 30 real queries	assignment brief (not in the zip), Section 8, Project 4: "at least 30 real queries"
E87	ICDT list "to be updated soon"	`supporting_files/A_collection/collection_log.md` section 7; `DATASET_CARD.md`
E88	52 / 3 commits	git history: `git log --format=%an | sort | uniq -c`
E89	A2 exports committed by myankb08	`git log --author=myankb08` (both `export_A2.json` commits); `LABELLING_PROTOCOL.md` section 9
E90	form shared by hem436	`supporting_files/C_labelling/forms_log.md`, row 2
E91	18 requests to the assistant	`prompts/assistant/user_requests.md`, table rows
Added with the section structure (2.1 to 7.4)		

id	what	file and field
E92	metadata fields: source, timestamp, collector, tool, tool version, content hash	`data/metadata.csv` header (`source_doc`, `source_page`, `collected_at_utc`, `collector`, `tool`, `tool_version`, `content_sha256`); `docs/DATA_CONTRACT.md` "Metadata"
E93	licences and use of institute documents and course plans	`docs/CONSENT_AND_LICENSING.md`, "Institute documents", "Course plans"
E94	consent register, one row per response	`data/consent_register.csv` (output of DVC stage `import_forms`), `consent_given`, `form_version`
E95	data flow and stage order	`docs/DATA_FLOW.md`; `dvc.yaml` stages
E96	collection log	`supporting_files/A_collection/collection_log.md`
E97	Project 4 checks V06, V11, V12, V13, V17 to V20, V25	`supporting_files/B_validation/processed/results.csv`, `check_id`, `description`
E98	label order decision, category, edit	`LABELLING_PROTOCOL.md` section 5; `labelstudio/config_v1.xml`
E99	dataset card: sources, sizes, balance, biases	`DATASET_CARD.md`, sections Sources, Sizes, Splits and balance, Known biases and limits
E100	100 eligible, 100 not eligible pairs	`supporting_files/B_engine/queries_summary.json`, `eligibility_eligible`, `eligibility_not_eligible`
E101	labelling protocol contents	`LABELLING_PROTOCOL.md`, sections 1 to 9
E102	README part map and rebuild commands	`README.md`, "Rebuild the data", "Where each part lives"
E103	5 gold items with "workshop track"	`data/gold/gold_test.jsonl`, `audited_query`; list in `HANDOFF_A2.md` errata

B AI assistance

What the assistant did. Claude Code (model claude-opus-5-5) wrote most of the skeleton code through subagents whose prompts are in `prompts/assistant/`: `agent_common_rules.md`, `agent_A_curriculum.md`, `agent_A2_ee_pages.md`, `agent_B_engine.md`, `agent_B2_engine_revision.md`, `agent_C_labelling.md`, `agent_D_validation.md`, `agent_E_llm_augment_al.md`, `agent_F_splits_dvc_docs.md`, `agent_F2_resplit.md`, `agent_G_report.md` and `agent_G2_report_redesign.md` (this report). hem436's own requests to the assistant are summarised, each with the files it changed, in `user_requests.md` [E91]. The assistant also drafted the 30 adjudication proposals (hem436 and myankb08 made every final call).

ChatGPT. hem436 used ChatGPT to reword 9 questions that classmates asked in person before entering them in the form (`prompts/assistant/chatgpt_form_rewording.md`).

Pipeline LLM. Qwen2.5-1.5B-Instruct (Q4_K_M via llama.cpp, or fp16 via transformers on Kaggle) paraphrases queries and justifications and gives base-model answers. Its prompts are files in `prompts/pipeline/`: `paraphrase_query_v1.txt`, `paraphrase_justification_v1.txt`, `augment_paraphrase_v1.txt` and `base_model_answer_v1.txt`. Model, digest and backend are stored with every output, and the README there lists model, versions and backends per prompt and the Qwen default system prompt.

What people did. All labelling (A1 and A2, both rounds), every adjudication decision, the scope and modelling decisions in `docs/DECISIONS.md`, both gold-bug findings, and form sharing and distribution.

C Guidelines v2 and label levels

Label levels, in this order: 1) decision: accept, edit or reject; 2) one error category, by priority: hallucinated course, wrong fact or policy, meaning drift, incomplete justification, ambiguous query, fluency (or none); 3) edit text, only for edit. The label set and the Label Studio config did not change between rounds.

Changes from v1 to v2, each tied to round-1 disagreements:

- R1: a justification must state the gold outcome and every number, course and open item; pure encouragement is an incomplete justification.
- R2: any number or status that contradicts gold or is not in gold is a wrong fact, including vague text that implies a wrong status.

- R3: edit when the fix touches at most two sentences or one query phrase, otherwise reject; a missing fact alone is never a reason to reject.
- R4: added scope, “available” for “remaining”, or a question turned back on the student is meaning drift.
- R5: with two problems, pick the category by priority and fix both texts.
- R6: gold is the truth; do not correct it from memory, write “GOLD?” instead.
- R7: formatting and tone are not errors; edit or reject always needs a category other than none.

D Detailed tables

Table 1: Failing records before and after cleaning. Source: `supporting_files/B_validation/before_after.md`.

check	what it tests	raw	after
V02	code format	6	0
V03	credits are integers in range	49	0
V05	required fields present	9	0
V06	no duplicate codes	5	0
V07	no exact duplicate records	4	0
V08	title matches timetable	5	0
V09	title spacing and punctuation	4	0
V10	near-duplicate titles (warning)	2	2

Table 2: Augmentation checks. Source: `supporting_files/D_augmentation/invariance.csv`.

augmentation	check	children	kept	rate
LLM paraphrase	gold and stated facts unchanged	120	36	0.30
typos and abbreviations	gold unchanged	600	600	1.00
transcript order	gold unchanged	445	445	1.00
transcript length	gold recomputed correctly	445	445	1.00

Table 3: Share of the 232 failures captured per budget, and coverage. Random is the mean of 5 seeds. Source: `supporting_files/D_active_learning/label_efficiency.csv`.

strategy	25	50	100	types	intents
top-k by disagreement	0.11	0.22	0.43	0.60	1
stratified by intent (added later)	0.10	0.19	0.38	1.00	3
random	0.08	0.15	0.30	0.88 to 0.92	3

Table 4: Split sizes. Students are those assigned; pairs per programme are CSE / AI / EE / ICDT. Source: `supporting_files/E_splits/split_summary.json`; shifted row from `supporting_files/E_splits/shifted_summary.json`.

split	students	pairs	children	rows	eligible / not	by programme
train	101	336	869	1205	56 / 56	127 / 90 / 119 / 0
val	24	80	200	280	13 / 13	35 / 21 / 24 / 0
test	31	102	251	353	17 / 17	49 / 21 / 32 / 0
ICDT holdout	24	82	206	288	14 / 14	0 / 0 / 0 / 82
shifted (forms)	n/a	19	0	19	n/a	n/a